

How Watson works

Watson is a workload optimised system based on IBM DeepQA architecture running on a cluster of IBM POWER7 processor-based servers. DeepQA is a massively parallel probabilistic evidence-based architecture. More than 100 different techniques are used to analyse natural language, identify sources, find and generate hypotheses, find and score evidence, and merge and rank hypotheses. Far more important than any particular technique is the way all these techniques are combined in DeepQA such that overlapping approaches can bring their strengths to bear and contribute to improvements in precision, confidence or speed.

Early implementations of Watson ran on a single processor, which required two hours to answer a single question. Of course, this was improved over time and in the final avatar, Watson housed 90 clustered IBM Power 750 servers. Each server features 32 POWER7 processors, running eight 3.55GHz cores each. As you can see, total power is staggering. IBM says the system has a combined total of 16 terabytes of memory and can operate at over 80 Teraflops (trillions of operations per second).

Of course, in addition to all this, Watson needed to be fed the ability to learn natural language processing. The super-computer has roughly 200 million pages of natural language content – equivalent to reading one million books.

The main innovation here, as you can see, isn't Watson's ability to come up with an answer to the proposed problem. It's the ability to process natural language in such a way that it can come up with the same answer in several different ways at lightning-fast speeds. The ability to reach the same answer over and over is where the "confidence" part of the challenge comes in – the most times it reaches the same conclusion, the more confident it is. And the speed is important since a human might be able to buzz in faster with a lower level of confidence than Watson.

Unlike conventional computing technologies designed to return documents containing the user's keywords or semantic entities, Watson is expected to leap ahead to interpret the user's query as a true question and to determine precisely what the user is asking for. Watson uses massively parallel processing to simultaneously and instantly understand complex questions – questions that require the system to consider huge volumes and varieties of natural language text to gather and then deeply analyse and score supporting or refuting evidence. The system then decides how confident it is in the answer. This approach marries advanced machine